

Control for Grid-Connected and Intentional Islanding Operations of Distributed Power Generation

Irvin J. Balaguer, *Student Member, IEEE*, Qin Lei, Shuitao Yang, Uthane Supatti, *Student Member, IEEE*, and Fang Zheng Peng, *Fellow, IEEE*

Abstract—Intentional islanding describes the condition in which a microgrid or a portion of the power grid, which consists of a load and a distributed generation (DG) system, is isolated from the remainder of the utility system. In this situation, it is important for the microgrid to continue to provide adequate power to the load. Under normal operation, each DG inverter system in the microgrid usually works in constant current control mode in order to provide a preset power to the main grid. When the microgrid is cut off from the main grid, each DG inverter system must detect this islanding situation and must switch to a voltage control mode. In this mode, the microgrid will provide a constant voltage to the local load. This paper describes a control strategy that is used to implement grid-connected and intentional-islanding operations of distributed power generation. This paper proposes an intelligent load-shedding algorithm for intentional islanding and an algorithm of synchronization for grid reconnection.

Index Terms—Distributed generation (DG), grid-connected operation, intentional-islanding operation, islanding detection, load shedding, synchronization.

I. INTRODUCTION

ISLANDING is a condition in which a microgrid or a portion of the power grid, which contains both load and distributed generation (DG), is isolated from the remainder of the utility system and continues to operate [1]–[4].

The disconnection of the DG once it is islanded is required by the IEEE Std. 929-2000 [5] and by the IEEE Std. 1547-2003 [6]. With the increasing competition among the power companies to secure more and more customers, the pressure to maintain a high degree of uninterrupted power service quality and reliability is felt by the utility companies [7], [8]. Thus, in a deregulated market environment, current practices of disconnecting the DG following a disturbance will no longer be a practical or reliable solution. As a result, the IEEE Std. 1547-2003 states, as one of its tasks for future consideration, the implementation of intentional islanding of DGs [6].

During the grid-connected operation, each DG system is usually operated to provide or inject preset power to the grid, which

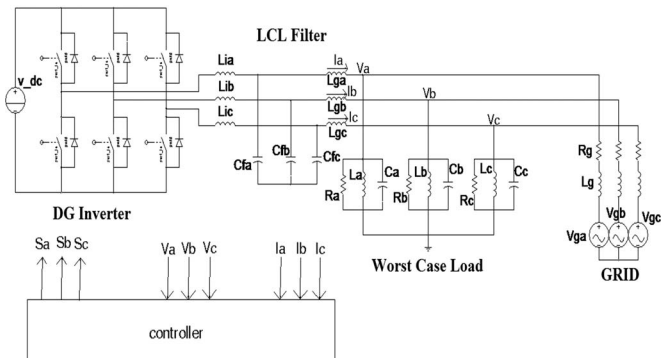


Fig. 1. Schematic diagram of the grid-connected inverter system.

is the current control mode in stiff synchronization with the grid [9]–[12]. When the microgrid is cut off from the main grid (intentional-islanding operation), each DG system has to detect this islanding situation and has to be switched to a voltage control mode to provide constant voltage to the local sensitive loads [13]–[15]. This paper describes a control strategy that is used to implement grid-connected and intentional-islanding operations of microgrids. The described method proposes two control algorithms, namely, one for grid-connected operations and the other for intentional-islanding operations. Specifically, this paper proposes an intelligent load-shedding algorithm for intentional islanding and an algorithm for synchronization for grid reconnection.

II. CONTROLLER

A. Introduction

Fig. 1 shows the main circuit topology. This system consists of the microsource that is represented by the dc source, the conversion unit which performs the interface function between the dc bus and the three-phase ac world, and the *LCL* filter that transports and distributes the energy to the end use and the load [16], [17]. The controller presented provides a constant DG output and maintains the voltage at the point of common coupling (PCC) before and after the grid is disconnected.

Under normal operation, each DG system in the microgrid usually works in a constant current control mode in order to provide a preset power to the main grid. When the microgrid is cut off from the main grid, each DG inverter system must detect this islanding situation and must switch to a voltage control mode. In this mode, the microgrid will provide a constant voltage to the local load.

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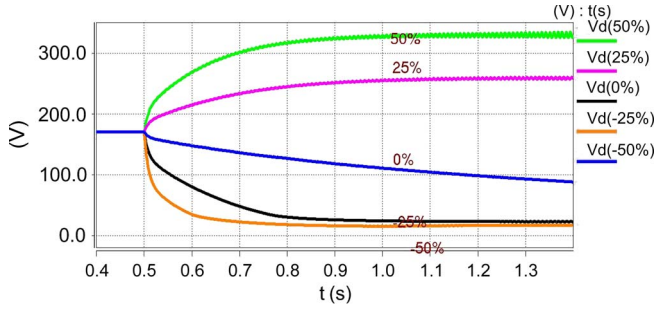


Fig. 5. Voltage transients under various active power differences.

implemented in a DG system to make sure that the demand is within available generation by disconnecting some least important loads [30].

D. Intelligent Load Shedding

Load shedding is defined as the process in which a part of the system load is disconnected according to a certain priority in order to steer the power system from potential dangers [31], [32]. During the grid-connected operation, the DG is operated to provide the optimum power to the grid according to many factors such as the availability of energy, energy cost, and so on [33]. The main grid is supplying or absorbing the power difference between the DG and the local load demand. When the main power grid is out (power outage), the DG that continues to inject predetermined optimum power can cause voltage and frequency transients, depending on the degree of power difference. The power difference makes the voltage and frequency drift away from the nominal values [34]. When the voltage and frequency drifts have reached certain levels, it is deemed that an islanding is occurring. This is the method that has been used to detect islanding. This methodology is enough for islanding detection. However, it is not enough for intentional-islanding operation, because often the local DG is either less or greater than the local load demand, and intelligent load shedding is needed. Therefore, it is essential to have an analytical solution of the voltage and frequency transients locally for the DG to have information and to make decisions and for intelligent load shedding to secure energy delivery to sensitive loads.

To develop the load-shedding algorithm, a constant impedance load is used. Fig. 5 shows the theoretical voltage transients for a constant impedance load under various active power differences (from -50% to $+50\%$) after main power outage, while Fig. 6 shows the theoretical frequency transients under various reactive power differences. As shown in Figs. 5 and 6, with no load shedding, it would be insufficient in keeping the voltage and frequency within the limits required.

When the voltage at the PCC has reached either less than 0.88 p.u. or beyond 1.1 p.u., the main power grid is deemed as an outage of service according to the IEEE Std. 1547 [6]. The challenge is how to switch the DG inverter system to the voltage control mode and how to bring the voltage back to the normal range (0.88–1.1 p.u.) for intentional-islanding operation. The analytical solution of the simple-case scenario shown in Fig. 5 provides a possible solution to this challenge. Fig. 5 shows that the voltage change rate is closely related to

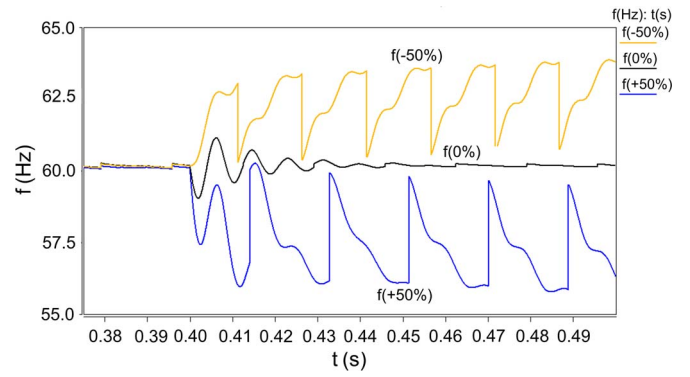


Fig. 6. Frequency transients under various reactive power.

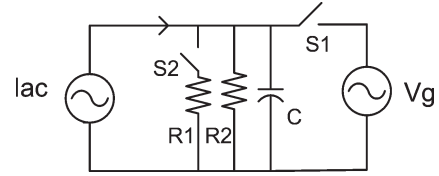


Fig. 7. System that is used to implement load shedding.

the power differences between the DG and the load demand. The approach that is proposed in this paper is used to detect the voltage change rate and profile after the power outage and to determine how much load shedding is needed before going to the intentional-islanding operation and switching to the voltage control mode. In order to accomplish this, the system that is shown in Fig. 7 has been analyzed.

To determine the amount of load that is to be disconnected, the following algorithm is proposed.

- 1) Obtain the voltage amplitude expression before load shedding. Using the circuit shown in Fig. 7, the expressions for the load voltages V_{apu} , V_{bpu} , and V_{cpu} can be found

$$V_{apu}(t) = \frac{I_{dpu} \cdot R_{2pu} \cdot Z_{Cpu}}{\sqrt{R_{2pu}^2 + Z_{Cpu}^2}} \cdot \sin(\omega t) \quad (5)$$

$$V_{bpu}(t) = \left(\frac{\frac{\sqrt{3}}{2} I_{dpu} \cdot R_{2pu} \cdot Z_{Cpu}}{\sqrt{R_{2pu}^2 + Z_{Cpu}^2}} \cdot e^{-\frac{Z_{Cpu}}{R_{2pu}} \omega t} + \left(\frac{-\frac{1}{2} I_{dpu} \cdot R_{2pu} \cdot Z_{Cpu}}{\sqrt{R_{2pu}^2 + Z_{Cpu}^2}} \cdot \sin(\omega t) \right) + \left(\frac{-\frac{\sqrt{3}}{2} I_{dpu} \cdot R_{2pu} \cdot Z_{Cpu}}{\sqrt{R_{2pu}^2 + Z_{Cpu}^2}} \cdot \cos(\omega t) \right) \right) \quad (6)$$

$$V_{cpu}(t) = \left(\frac{-\frac{\sqrt{3}}{2} I_{dpu} \cdot R_{2pu} \cdot Z_{Cpu}}{\sqrt{R_{2pu}^2 + Z_{Cpu}^2}} \cdot e^{-\frac{Z_{Cpu}}{R_{2pu}} \omega t} + \left(\frac{-\frac{1}{2} I_{dpu} \cdot R_{2pu} \cdot Z_{Cpu}}{\sqrt{R_{2pu}^2 + Z_{Cpu}^2}} \cdot \sin(\omega t) \right) + \left(\frac{\frac{\sqrt{3}}{2} I_{dpu} \cdot R_{2pu} \cdot Z_{Cpu}}{\sqrt{R_{2pu}^2 + Z_{Cpu}^2}} \cdot \cos(\omega t) \right) \right) \quad (7)$$

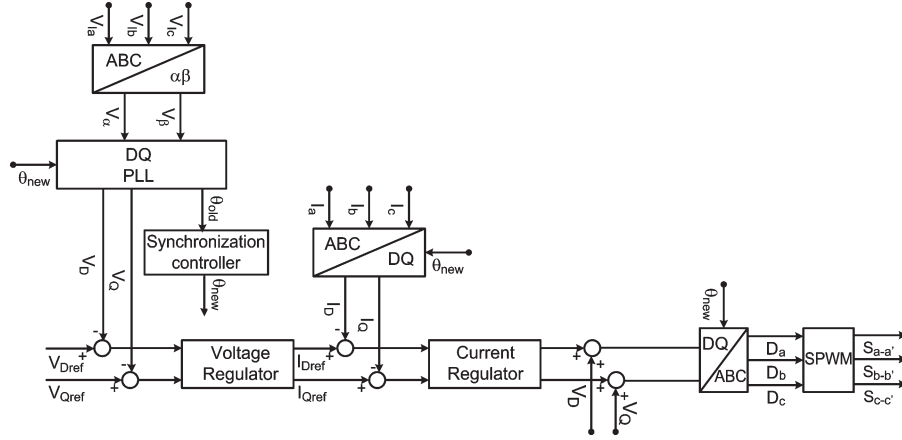


Fig. 8. Block diagram of the voltage-controlled inverter.

Using V_{apu} , V_{bpu} , and V_{cpu} , expressions for the voltage amplitude can be found at the bottom of the page as (8) and (9). If $K = (\omega t Z_{Cpu}/R_{2pu})$, then

$$\begin{aligned} V_d(t) &= 1 + \Delta V_d(t) \\ &= 1 + I_{dpu} R_{2pu} Z_{Cpu} \\ &\quad \times \sqrt{\frac{e^{-2K} (1 + e^{2K} - 2e^K \cdot \cos(\omega t))}{R_{2pu}^2 + Z_{Cpu}^2}}. \end{aligned} \quad (10)$$

- 2) Derive the slope of the voltage amplitude, which is shown at the bottom of the page as (11).
- 3) Derive I_{dpu} at a fixed time t_0

I_{dpu}

$$\begin{aligned} &= \frac{s \sqrt{(1 + e^{-2K} - 2e^{-K} \cos(\omega t)) (R_{2pu}^2 + Z_{Cpu}^2)}}{e^{-2K} \omega Z_{Cpu} (e^K \sin(\omega t) R_{2pu} + (-1 + e^K \cos(\omega t)) Z_{Cpu})}. \end{aligned} \quad (12)$$

- 4) Obtain the value of load to be shed

$$R_{2pu} = \frac{1}{\sqrt{I_{dpu}^2 - \frac{1}{Z_{Cpu}^2}}} \quad (13)$$

$$R_{1pu} = \frac{R_{Tpu} R_{2pu}}{R_{Tpu} - R_{2pu}} \quad (14)$$

where $R_{Tpu} = R_{1pu}/R_{2pu}$.

E. Intentional-Islanding Operation Mode

The voltage closed-loop control for intentional-islanding operation is shown in Fig. 8. The control works as voltage regulation through current compensation. The controller uses voltage compensators to generate current references for current regulation.

As shown, the load voltages (V_d and V_q) are forced to track its reference by using a PI compensator (voltage regulator). The outputs of this compensator (I_{dref} and I_{qref}) are compared with the load current (I_d and I_q), and the error is fed to a current regulator (PI controller). The output of the current compensator acts as the voltage reference signal that is fed

$$\Delta V_d(t) = I_{dpu} R_{2pu} Z_{Cpu} \sqrt{\frac{e^{-\frac{2\omega t Z_{Cpu}}{R_{2pu}}} \left(1 + e^{\frac{2\omega t Z_{Cpu}}{R_{2pu}}} - 2e^{-\frac{\omega t Z_{Cpu}}{R_{2pu}}} \cdot \cos(\omega t) \right)}{R_{2pu}^2 + Z_{Cpu}^2}} \quad (8)$$

$$V_d(t) = 1 + \Delta V_d(t) = 1 + I_{dpu} R_{2pu} Z_{Cpu} \sqrt{\frac{e^{-\frac{2\omega t Z_{Cpu}}{R_{2pu}}} \left(1 + e^{\frac{2\omega t Z_{Cpu}}{R_{2pu}}} - 2e^{-\frac{\omega t Z_{Cpu}}{R_{2pu}}} \cdot \cos(\omega t) \right)}{R_{2pu}^2 + Z_{Cpu}^2}} \quad (9)$$

$$s = \frac{dV_d(t)}{dt} = \frac{d\Delta V_d(t)}{dt} = \frac{e^{-2K} \omega I_{dpu} Z_{Cpu} (e^K \sin(\omega t) R_{2pu} + (-1 + e^K \cos(\omega t)) Z_{Cpu})}{\sqrt{(1 + e^{-2K} - 2e^{-K} \cos(\omega t)) (R_{2pu}^2 + Z_{Cpu}^2)}} \quad (11)$$

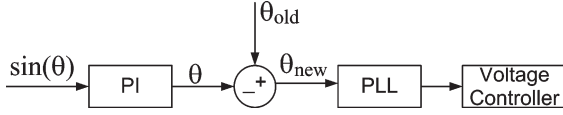


Fig. 9. Synchronization controller.

to the sinusoidal pulsewidth modulator to generate the high-frequency gating signals for driving the three-phase voltage source inverter. The current loop is included to stabilize the system and to improve the system dynamic response by rapidly compensating for near-future variations in the load voltages [35]. In order to get a good dynamic response, V_{DQ} is fed forward. This is done because the terminal voltage of the inverter is treated as a disturbance, and the feedforward is used to compensate for it [12].

F. Synchronization for Grid Reconnection

When the grid-disconnection cause disappears, the transition from islanded to grid-connected mode can be started. To avoid hard transients in the reconnection, the DG has to be synchronized with the grid voltage [36]–[38]. The DG is operated in the synchronous island mode until both systems are synchronized. Once the voltage in the DG is synchronized with the utility voltage, the DG is reconnected to the grid, and the controller will pass from the voltage to the current control mode. This synchronization is achieved by implementing the following algorithm.

- 1) Assume that the phase difference between the grid and inverter voltages is given by

$$\theta = \angle V_G - \angle V_I. \quad (15)$$

- 2) In order to obtain the information of θ , two sets of voltage values are used

$$\begin{aligned} k &= V_{Ia}V_{Ga} + V_{Ib}V_{Gb} + V_{Ic}V_{Gc} \\ &= \frac{3}{2} [\cos(\theta)] \end{aligned} \quad (16)$$

$$\begin{aligned} g &= V_{Ia}V_{Gb} + V_{Ib}V_{Gc} + V_{Ic}V_{Ga} \\ &= \frac{3}{4} [-\cos(\theta) + \sqrt{3}\sin(\theta)]. \end{aligned} \quad (17)$$

Using the variables k and g , $\sin(\theta)$ can be found as

$$\sin(\theta) = \frac{\frac{4}{3}g + \frac{2}{3}k}{\sqrt{3}}. \quad (18)$$

Fig. 9 shows how $\sin(\theta)$ is used to obtain the new phase angle for which the grid and inverter voltages are synchronized.

III. CONTROL ANALYSIS AND STABILITY

As previously mentioned, the control method used has two modes of control operation: current and voltage controls. These control modes correspond to the system's operating mode (grid connected or islanding, respectively). In order to determine the stability of these two controllers, their transfer functions have to be determined.

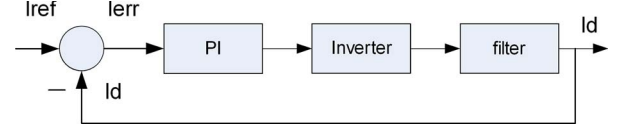
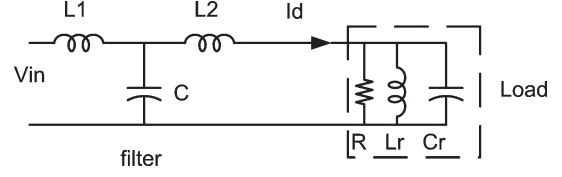


Fig. 10. Block diagram of the current-controlled inverter.

Fig. 11. LCL filter and parallel RLC load.

A. Current Control Transfer Function

Fig. 10 shows the block diagram of the DG interface control for the grid-connected operation.

The PI controller produces a signal that is proportional to the time integral of the controller. The transfer function of the PI controller is given by

$$C(s) = k_P + \frac{k_I}{s} \quad (19)$$

where k_p is the proportional gain and k_I is the integral gain.

The inverter stage does not have any significant transient time associated with it, and hence, it is modeled as an ideal gain. This ideal gain can be given by $G_I(s) = 1$.

The schematic circuit of the filter stage is shown in Fig. 11. It consists of an LCL filter and a parallel RLC load. The transfer function of this stage can be expressed as

$$\frac{I_d}{V_{in}} = \frac{\frac{1}{sC}}{\left(\frac{1}{sC} + sL_2 + R // sL_r // \frac{1}{sC_r}\right) \cdot Z_{total}} \quad (20)$$

where

$$Z_{total} = sL_1 + \frac{1}{sC} // \left(sL_2 + R // sL_r // \frac{1}{sC_r}\right). \quad (21)$$

Using (19), (20), and (21), the transfer function of the current-controlled system is given by (22), which is shown at the bottom of the next page.

It can be seen in (22) that the system is stable according to the conventional control theory. Fig 12 shows the Bode plot of the current-controlled inverter. As can be noticed, the system is stable with a positive phase margin.

B. Voltage Control Transfer Function

The voltage closed-loop control for intentional-islanding operation is shown in Fig. 13. The transfer function of this voltage controller system is given by (23), which is shown at the bottom of the next page.

It can be seen in (23) that the system is stable according to the conventional control theory. Fig. 14 shows the Bode plot of the current-controlled inverter. As can be noticed, the system is stable with a positive phase margin.

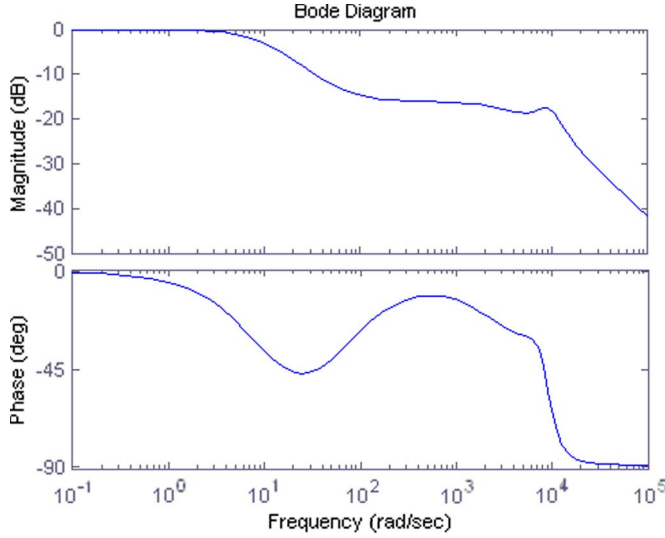


Fig. 12. Bode plot for the current-controlled inverter.

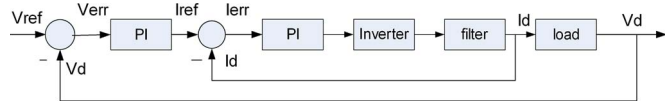


Fig. 13. Block diagram of the voltage-controlled inverter.

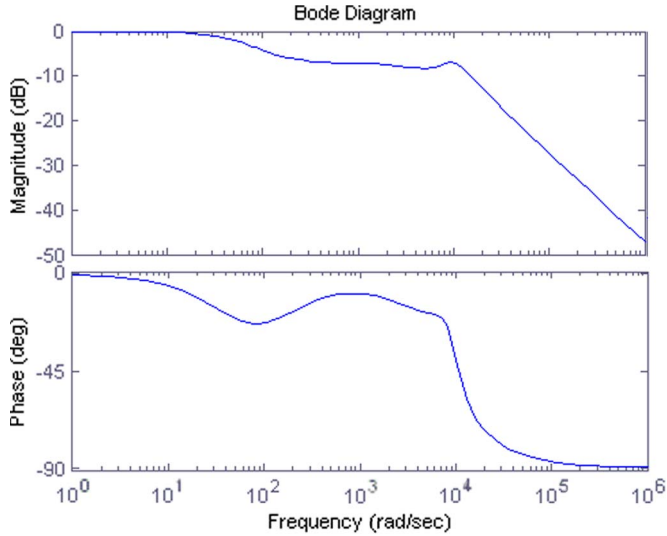


Fig. 14. Bode plot for the voltage-controlled inverter.

IV. SIMULATION RESULTS

The performance of the proposed control strategies was evaluated by computer simulation using SABER. Fig. 15 shows the simulated system. This system was tested under the following conditions:

- 1) switching frequency f_s : 10 kHz;
- 2) output frequency: 60 Hz;
- 3) filter inductor L_i : 1 mH;
- 4) filter inductor L_L : 0.5 mH;
- 5) filter capacitor C_f : 31 μ F;
- 6) dc-link voltage V_{dc} : 400 V;
- 7) output phase voltage $V_{o1\phi}$: 120 Vrms;
- 8) output capacity: 10 KW.

The RLC load was adjusted to be resonant at 60 Hz and to consume 10 KW. The DG system was designed to supply 10 KW and zero reactive power. The system was operated initially in grid-connected operation. The grid was disconnected at 0.3 s, and this event was detected at 0.30155 s. After 0.30155 s, the control mode was changed from current- to voltage-controlled operation. Fig. 16 shows the voltages and currents at the PCC before and after grid disconnection.

The grid was reconnected at 0.6 s. The DG was operated in the synchronous island mode until both systems were resynchronized. Fig. 17 shows the synchronization of the voltages at both ends of the PCC when the synchronization algorithm starts to work in the intentional-islanding mode. As can be seen, the proposed algorithm successfully forces the voltage at the DG to track the voltage at the grid.

Once the synchronization was completed, the DG was reconnected to the grid, and the controller was switched from the voltage to the current control mode. Fig. 18 shows the phase voltage V_a without and with the synchronization algorithm implemented. Notice that the algorithm avoids a hard transient in the reconnection from intentional-islanding to grid-connected operation.

To keep the magnitude of the voltage in its normal operational range when there is a power mismatch, the load-shedding algorithm proposed was implemented. Fig. 19 shows the theoretical voltage transients under a power difference of -50% , without the load-shedding algorithm implemented. For this case, when the voltage is out of the normal operating point, the load-shedding algorithm cuts off the power difference from the load, and the voltage was brought back to the

$$T(s) = \frac{s^3 + 8.72 \times 10^3 s^2 + 6.51 \times 10^7 s + 4.03 \times 10^9}{s^4 + 9.46 \times 10^3 s^3 + 1.04 \times 10^8 s^2 + 3.31 \times 10^{11} s + 3.22 \times 10^{12}}$$

$$(L_1 = 1 \text{ mH}, L_2 = 0.5 \text{ mH}, C = 31 \mu\text{F}, R = 4.33, L_r = 4.584 \text{ mH}, C_r = 1.535 \text{ mF}, k_P = 0.8, k_I = 50) \quad (22)$$

$$T(s) = \frac{s^4 + 8.79 \times 10^3 s^3 + 6.56 \times 10^7 s^2 + 8.06 \times 10^9 s + 6.45 \times 10^7}{s^5 + 1.42 \times 10^4 s^4 + 1.46 \times 10^8 s^3 + 6.44 \times 10^{11} s^2 + 3.49 \times 10^{13} s + 2.79 \times 10^{11}}$$

$$(L_1 = 1 \text{ mH}, L_2 = 0.5 \text{ mH}, C = 31 \mu\text{F}, R = 4.33, L_r = 4.584 \text{ mH},$$

$$C_r = 1.535 \text{ mF}, k_{P1} = 0.8, k_{I1} = 50, k_{P2} = 1.24, k_{I2} = 0.02) \quad (23)$$

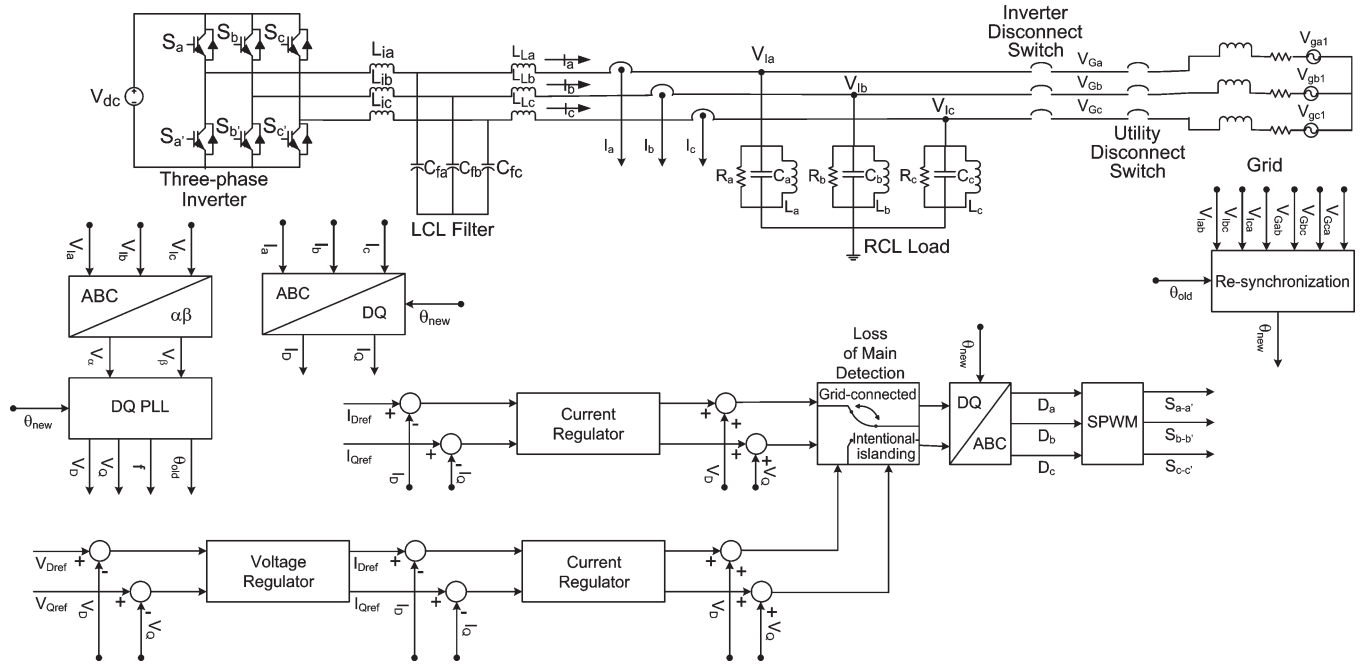


Fig. 15. Simulated system.

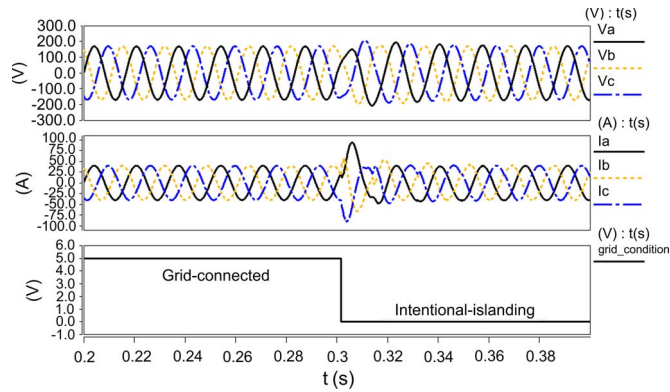


Fig. 16. From grid-connected to intentional-islanding operation.

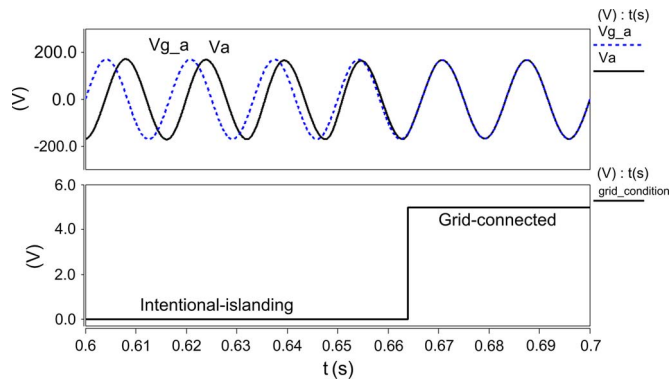


Fig. 17. Synchronization for grid reconnection.

normal range. Fig. 20 shows that the suitable load disconnection results in voltage recovery, compared to the case of “no load shedding.”

The proposed control strategy was evaluated with two DGs connected in parallel, forming a microgrid, as shown in Fig. 21.

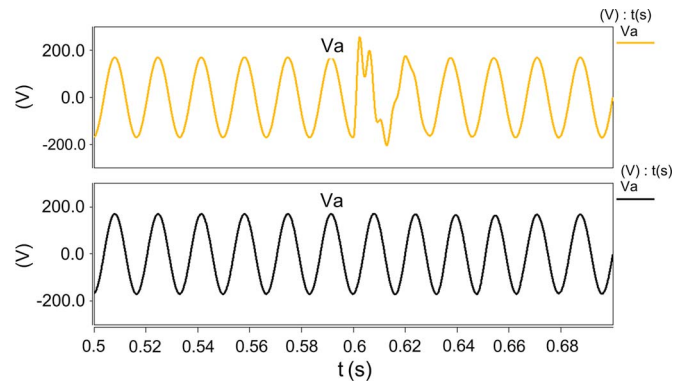
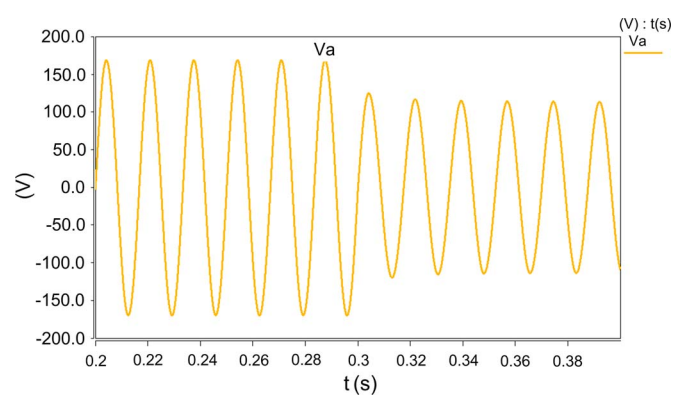


Fig. 18. Phase voltage (top) without and (bottom) with the synchronization algorithm.

Fig. 19. Phase voltage V_a without the load-shedding algorithm.

DG1 was controlled as a constant current control when the grid was connected to the system and as a constant voltage control when the grid was disconnected (intentional islanding). DG2 was controlled as a constant current control all the time (grid-connected and intentional-islanding operations).

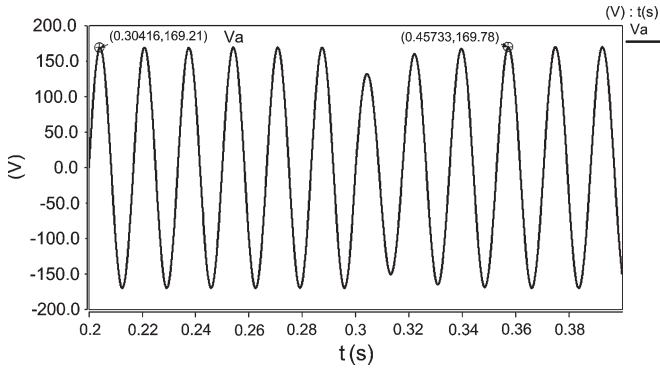
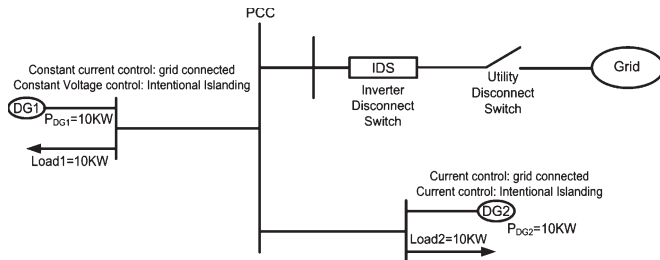
Fig. 20. Phase voltage V_a with the load-shedding algorithm.

Fig. 21. Microgrid configuration.

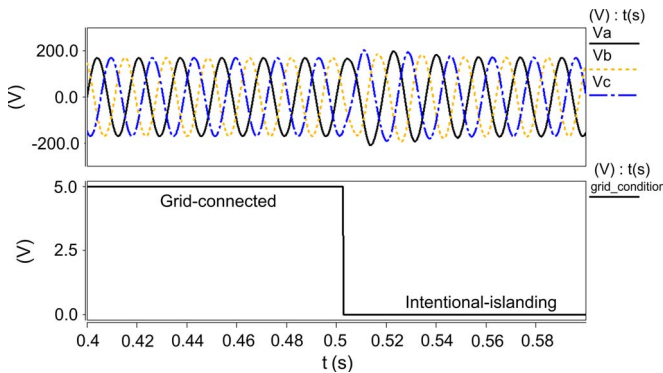


Fig. 22. Microgrid voltages: from grid connected to intentional islanding.

Both RLC loads were adjusted to be resonant at 60 Hz, and they consume 10 KW. Each DG system was designed to supply 10 KW and zero reactive power. The system was operated initially in grid-connected operation. The grid was disconnected at 0.5 s, and this event was detected at 0.50256 s. After 0.50256 s, the control mode of DG1 was changed from current-to-voltage-controlled operation, while the control mode of DG2 was kept as a constant current control. Fig. 22 shows the voltages at the PCC before and after grid disconnection.

The grid was reconnected at 1 s. Both DGs were operated in the synchronous island mode until both systems were resynchronized. Fig. 23 shows the synchronization of the voltages at both ends of the PCC when the synchronization algorithm starts to work in the intentional-islanding mode. As can be seen, the proposed algorithm successfully forces the voltage at the microgrid to track the voltage at the grid. Once the synchronization was completed, the microgrid was reconnected to the grid, and the controller for DG1 was switched from the voltage to the current control mode.

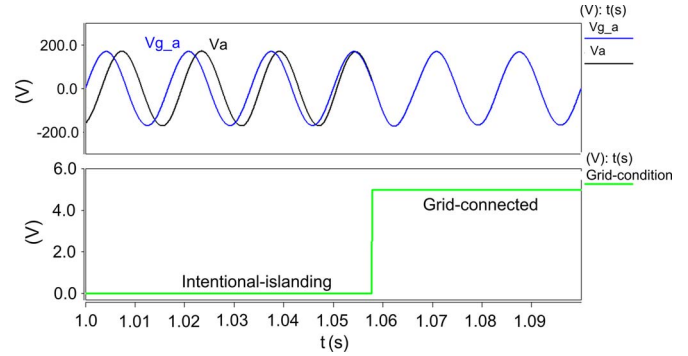


Fig. 23. Synchronization for grid reconnection (two DGs).

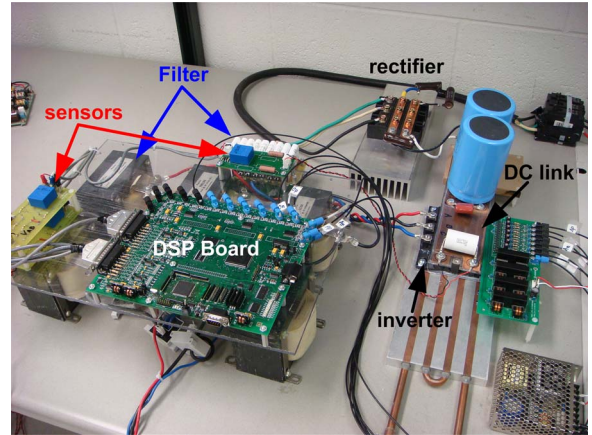


Fig. 24. Experimental setup.

V. EXPERIMENTAL RESULTS

The hardware prototype of Fig. 1 has been implemented for experimental verification. The control, PLL, grid condition detection, and reclosure algorithms have been programmed using a universal DSP control board developed at the Power Electronics and Motor Drives Laboratory, Michigan State University. The system was tested under the following conditions to experimentally verify the simulation results:

- 1) switching frequency f_s : 10 kHz;
- 2) output frequency: 60 Hz;
- 3) dead time: 3 μ s;
- 4) filter inductor L_i : 1 mH;
- 5) filter inductor L_L : 0.5 mH;
- 6) filter capacitor C_f : 50 μ F;
- 7) simulated output voltage: 104 V_{RMS-LL} and 3 Φ @ 60-Hz grid connection, with $V_{dc} = 200$ V;
- 8) output capacity: 2.5 KW.

The reason for simulating the utility voltage is to ensure that the algorithms and controllers are functioning properly under low-power tests, such that there is a reduced risk of operator and equipment damage if the system fails.

Shown in Fig. 24 are the inverter, the DSP board, the filter, and the rectifier.

A. Transition From Grid-Connected to Intentional-Islanding Operation

The DG is started up in the grid-connected operation mode, and then, the separation device is opened. When the DG is

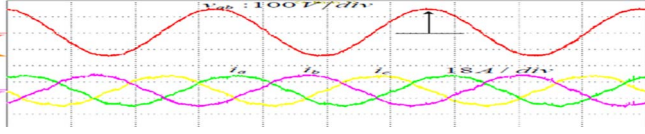


Fig. 25. Line-to-line voltage and phase currents during grid connected.

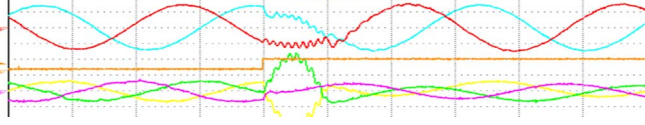


Fig. 26. Transition from grid-connected to intentional-islanding operation. (Top) Voltages. (Bottom) Currents.

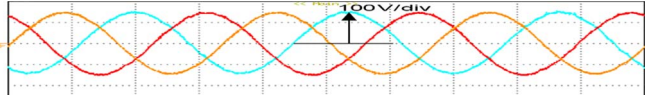


Fig. 27. Line-to-line voltage during the intentional-islanding operation.

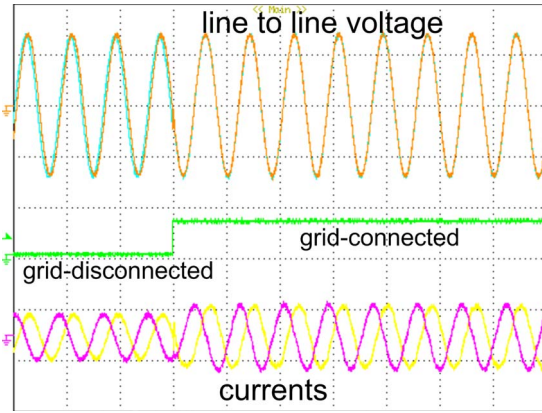


Fig. 28. Transition from intentional-islanding to grid-connected operation.

disconnected from the grid, it operates in the intentional-islanding mode. Fig. 25 shows how the system line-to-line voltage and phase current behave during the grid-connected mode.

Fig. 26 shows the corresponding line-to-line voltage and phase current when the disconnection device is opened.

B. Transition From Intentional-Islanding to Grid-Connected Operation

Fig. 27 shows the line-to-line voltage when the system is operating in the islanding mode. As can be seen, the proposed control scheme is capable of maintaining the voltages within the designed levels.

Fig. 28 shows the process of synchronization, where the line-to-line voltage at both ends of the separation device is illustrated. At the beginning of the synchronization, both voltages are out of phase. As can be seen, the proposed algorithm successfully forces the voltage at the DG to track the voltage at the grid until the synchronization process is completed. Also, shown is the smooth transition of the currents.

C. Load Shedding

The test case analyzed shows a situation where the islanded network is supplying 330 W and importing 330 W of active

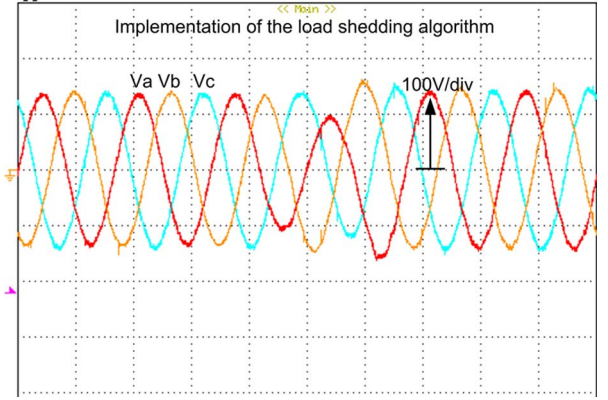


Fig. 29. Implementation of the load-shedding algorithm.

power from the grid in order to be able to supply the total load and to keep the load voltage at 80 Vrms. Starting from this point, in steady state, the DG is disconnected, and the network will become islanded. As shown in Fig. 29, it can be noticed that the suitable load disconnection results in voltage recovery, compared to the case of “no load shedding.” A total load of around 640 W is curtailed to 320 W through load shedding, which is within the DG capabilities. It can also be noticed from Fig. 29 that the load shedding assists the voltage to reach acceptable values above the threshold selected.

VI. CONCLUSION

Through this paper, the control, islanding detection, load shedding, and reclosure algorithms have been proposed for the operation of grid-connected and intentional-islanding DGs.

A controller was designed with two interface controls: one for grid-connected operation and the other for intentional-islanding operation. An islanding-detection algorithm, which was responsible for the switch between the two controllers, was presented. The simulation results showed that the detection algorithm can distinguish between islanding events and changes in the loads and can apply the load-shedding algorithms when needed. The reclosure algorithm causes the DG to resynchronize itself with the grid. In addition, it is shown that the response of the proposed control schemes is capable of maintaining the voltages and currents within permissible levels during grid-connected and islanding operation modes. The experimental results showed that the proposed control schemes are capable of maintaining the voltages within the standard permissible levels during grid-connected and islanding operation modes. In addition, it was shown that the reclosure algorithm causes the DG to resynchronize itself with the grid.

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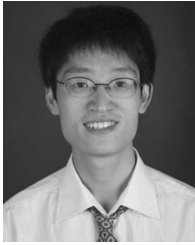
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